Another nonisotopic point configuration, of 21 points, is explicitly constructed in White [559]. A general construction method is due to Mnëv [408]. In particular, Mnëv's universality theorem (see below) shows that the "realization space" for planar point configuration can be arbitrarily complicated.

Suvorov [537] and Jaggi et al. [290] furthermore construct non-isotopic configurations whose points are in general position. From these, one can — using a technique of Sturmfels [117, Thm. 6.5] — get *simplicial* polytopes that violate the isotopy conjecture. This yields examples for a much more general "universality theorem for polytopes," described in the next section.

6.6 Rigidity and Universality

We have characterized Gale diagrams of polytopes in Corollary 6.20. To make statements about polytopes of a fixed combinatorial type, however, it is not sufficient to look at a specific Gale diagram: we have to make sure that the statement we make holds for *all polytopes combinatorially equivalent to the given polytope.*

Here we have to deal with two different notions of equivalence. We have noted that two polytopes may be combinatorially equivalent but have different oriented matroids. (See, for example, the octahedra of Example 6.3.) However, any two polytopes with the same oriented matroid are combinatorially equivalent: they have the same covectors, hence the same positive covectors, and hence the same cofaces, and thus the same faces.

To make Gale diagrams useful for high-dimensional polytopes, we have to get a hold on all Gale diagrams representing a combinatorial equivalence class of polytopes. This is not simple, and it would lead into a discussion of "partial oriented matroids" that we want to avoid (there is not much theory for that, either). Instead, we restrict ourselves to an important special case: when only one oriented matroid is possible for a given convex polytope.

Definition 6.25. The oriented matroid of a d-polytope $P \subseteq \mathbb{R}^d$ is *rigid* if every polytope P' that is combinatorially equivalent to P has the same oriented matroid.

Here our convention is to identify the vertex sets of P and of P' with $[n]$ in a way that is compatible with the combinatorial equivalence. With this, *combinatorial equivalence* means that the oriented matroids $\mathcal{M}(P)$ and $\mathcal{M}(P')$ have the same set of *positive cocircuits*, and *rigidity* means that this implies that *all* cocircuits of P and of P' coincide:

$$\mathcal{C}^*(P) \cap \{+,0\}^n = \mathcal{C}^*(P') \cap \{+,0\}^n \quad \Longrightarrow \quad \mathcal{C}^*(P) = \mathcal{C}^*(P').$$

For example, triangular prisms are rigid, but octahedra are not. The concept of rigidity is only interesting because there are some rigid polytopes around, although "most" polytopes are not rigid. Here is one construction to get rigid polytopes.

Theorem and Definition 6.26 (The "Lawrence construction"). *Let $V \in \mathbb{R}^{r\times n}$ be a vector configuration in $\mathbb{R}^r$, possibly obtained from a point configuration $X \in \mathbb{R}^{d\times n}$ in $\mathbb{R}^d$ with $r = d+1$. (To avoid trouble, we assume V has no coloops: even if we delete one of its vectors, the others still span $\mathbb{R}^r$.)*

If $G \in \mathbb{R}^{n\times(n-r)}$ is a Gale diagram of V, then adding the opposite to every vector of G we get the Gale diagram

$$\widehat{G} := \begin{pmatrix} G \\ -G \end{pmatrix} \in \mathbb{R}^{2n\times(n-r)}$$

of a polytope with $2n$ vertices in $\mathbb{R}^{2n-(n-r)-1} = \mathbb{R}^{n+d}$; this polytope is denoted by $\Lambda(V) \subseteq \mathbb{R}^{n+d}$ and called the Lawrence polytope of V.

Equivalently, we get the Lawrence polytope $\Lambda(V)$ by successive Lawrence extensions $V \longrightarrow \sigma_i(V)$: for this we replace each vector "$\boldsymbol{v}_i$" in V by two new vectors

$$\boldsymbol{v}_{i^+} \ := \ \boldsymbol{v}_i + \boldsymbol{e}_{r+i}$$

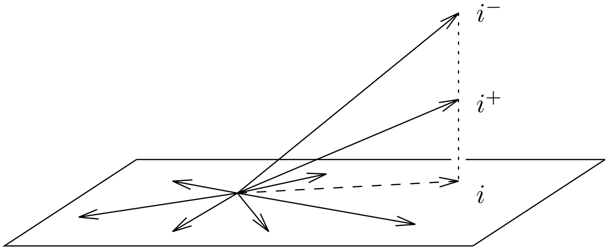
and

$$\boldsymbol{v}_{i^-} \ := \ \boldsymbol{v}_i + 2\boldsymbol{e}_{r+i},$$

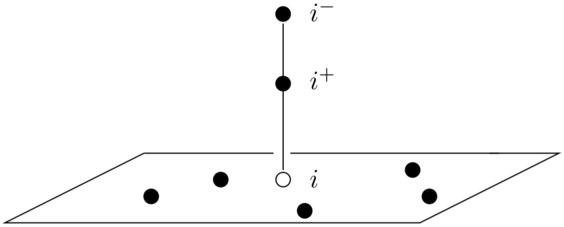
to get an acyclic vector configuration in $\mathbb{R}^{r+n}$, from which we pass to affine space $\mathbb{R}^{d+n}$.

If we start with an affine point configuration X, we can perform the Lawrence liftings directly on the point configuration, without linearization.

Our figures illustrate both the "linear picture" of a single Lawrence lifting applied to a vector configuration

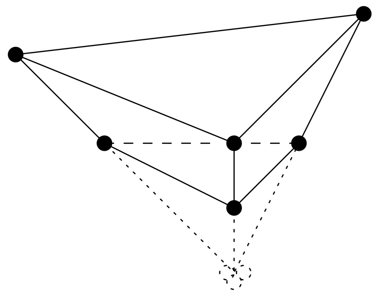


and the "affine picture," where a Lawrence lifting is performed on a single point in a finite point configuration, thus increasing the dimension of the configuration by 1:



Proof. It is easy to check that the Lawrence construction in fact yields a convex polytope and that the two descriptions are equivalent. ☐

As a very trivial example, consider three nonzero vectors in $\mathbb{R}^1$ ($r = 1$, $n = 3$). The Lawrence construction applied to them yields a triangular prism in $\mathbb{R}^3$: we show the affine picture for this.



Theorem 6.27. *Lawrence polytopes are rigid, that is, if P' is combinatorially equivalent to $\Lambda(V)$, then the oriented matroid of P' is also isomorphic to that of P. In particular, the Gale diagrams of P' and of $\Lambda(V)$ are isomorphic.*

Proof. In the Gale diagram G of the Lawrence polytope $\Lambda(V)$, and in every Gale diagram G' with the same positive circuits, the points come in pairs of positive and negative points, since those pairs form positive circuits.

Furthermore, if we take any other circuit, then it contains at most one point from every such pair. Hence we get all the circuits from the positive ones by replacing a positive point by the negative "other point" of its pair. Thus all circuits of the diagram are determined by the positive ones, and thus the configuration V is rigid. ☐

The Lawrence construction has numerous applications. Perhaps the most striking one is the "universality theorem" for polytopes, which we want to describe now.

For this, one needs a suitable equivalence relation for semialgebraic sets; we use Richter-Gebert's version from [459]. Two semialgebraic sets S and T are *stably equivalent* if they can be related by a sequence of "rational changes of coordinates" (such that f and f^{-1} are both rational functions with $\mathbb{Q}$-coefficients, and induce homeomorphisms of the sets that we consider) and "stable projections" (whose fibers are the relative interiors of rational polyhedra) — see Richter-Gebert [459, Sect. 2.5] for the precise definitions and more details. Stable equivalence is a very "restrictive" concept. In fact, stably equivalent sets S and T

- have the same homotopy type (in particular, S is connected if and only if T is connected)

- have the same algebraic complexity (in particular, S contains rational points if and only if T has rational points),
- have comparable singularity structure (in particular, S is a manifold if and only if T is a manifold.

Universality Theorem for Polytopes 6.28. (Mnëv [408])
Every elementary semialgebraic set defined over $\mathbb{Z}$ is stably equivalent to the realization space of some polytope.

Every open elementary semialgebraic set defined over $\mathbb{Z}$ is stably equivalent to the realization space of some simplicial polytope.

Essentially, this means that the realization space of a polytope can be "arbitrarily complicated": it can be disconnected with many components, it can consist of circles and spheres (can have "homology" in all dimensions), and can have all kinds of complicated singularities — in general it is certainly not a manifold, as claimed in [465, p. 18].

The theorem on which all of this is based is *Mnëv's universality theorem*: the realization space for planar point configurations (i.e., for oriented matroids of rank 3) can be any semialgebraic set, up to stable equivalence.

For a long time, there was no detailed proof available for this theorem. Mnëv's paper [408] only provides a sketch of the basic ideas for the "local" version of the theorem; two further sketches are in Shor [499, Sect. 4] and in Goodman & Pollack [237, Sect. 7]; see also Björner et al. [96, Sect. 8.6]. Finally, a complete, detailed proof was provided by Günzel [261]. His proof also covers the far-reaching extension announced in Mnëv [409], the "universal partition theorem" for oriented matroids.

On the other hand, it is much easier to see (using the "van Staudt constructions" for addition and multiplication of points, of classical projective geometry [276, Sect. VI.7] [118, Sect. 2.1] [558, Sect. 7]) that the smallest subfield of $\mathbb{R}$ over which all planar point configurations can be realized is the field of all algebraic numbers $\mathbf{A} \subseteq \mathbb{R}$. This means with Theorem 6.28 that $\mathbf{A}$ is also the smallest field over which all polytopes can be realized.

There is a great new development: Richter-Gebert's Universality Theorem for 4-Polytopes, and the (even stronger) Universal Partition Theorem for 4-Polytopes, with all their corollaries and extensions.

Universality Theorem for 4-Polytopes 6.29. (Richter-Gebert [459])
Every elementary semialgebraic set defined over $\mathbb{Z}$ is stably equivalent to the realization space of some 4*-dimensional polytope.*

In the course of this work — done after the first version of this book appeared — Richter-Gebert solved quite a number of basic open problems (see Problems 5.11*, 6.10*, and 6.11*). There is neither time nor space to explain Richter-Gebert's work [459] here (see also Günzel [262]); an announcement appeared as [462], a survey is Richter-Gebert [460].

Notes

For all information about oriented matroids, we rely on the monograph by Björner et al. [96]. Other expositions that include surveys on oriented matroids are Bachem [34], Bachem & Kern [35], Bokowski & Sturmfels [118], and Bokowski [112].

As you may have noticed, we have deliberately tried to keep linear algebra concepts low-key. You may reformulate all of the basic constructions in more advanced language. For that, the vector configuration $V \in \mathbb{R}^{r \times n}$ is considered as a linear map $V : \mathbb{R}^n \longrightarrow \mathbb{R}^r$, the space $\mathrm{Dep}(V)$ is the kernel of this map, $\mathrm{Val}(V)$ is the image of the dual map, and so forth.

Deletion and *contraction* of an "element" are fundamental operations in many areas: for graphs (see Section 4.1), for vector configurations and oriented matroids (see Section 6.3(d)), and for arrangements and zonotopes (see the next lecture). In fact, there is a tremendous power in proofs "by deletion and contraction," which proceed by induction on the number of elements, and by putting a structure together from the information given by deletion and contraction of the same element. Zaslavsky's work on hyperplane arrangements [572] is the classic source for that approach.

Gale diagrams are a tool that emerged from work of Gale [220] and were developed to their full power and beauty by Perles, as documented in Grünbaum's book [252]. Additional sources are the book by McMullen & Shephard [403, Ch. 3], McMullen's survey [394], and the treatment (with nice illustrations and examples) in Ewald's book [201]. See Eisenbud & Popescu [195] for an algebraic geometry perspective. It seems that the close connection between the Gale diagram technique and oriented matroid duality was first mentioned in [394], and the explicit identification was worked out by Sturmfels [531].

The reduction to affine Gale diagrams is implicit in Perles' work (see Grünbaum [252, p. 59]) and also used by Bokowski [116]; it appears as a tool of its own standing in Sturmfels' work [532]. We mention for completeness that any two affine Gale diagrams of the same polytope are connected by a projective transformation and the corresponding reorientation.

Many interesting properties of polytopes can be profitably studied from the oriented matroid point of view, not only via Gale diagrams. Surveys of applications are in Grünbaum's book [252], in Bokowski & Sturmfels [118], and in Bayer & Lee [63, Sect. 4].

Some authors distinguish between "Gale diagrams" and "Gale transforms." We did not make this distinction, but essentially what we constructed here were Gale transforms, while any configuration that represents the dual oriented matroid is a Gale diagram. We also just note that there are several useful reformulations and variations of the Gale diagram construction, among them a "coordinate-free" formulation [202] [394], which was useful in the investigation of infinite-dimensional polytopes by Kleinschmidt & Wood [338, 569].